

# Weekly Work Plan: Met Gala 2026

## 1. Project Planning Week 1 (May 4 – May 10)

- Topic agreed on May 5: Met Gala 2026 social media discussion analysis.
- Data source: YouTube comments (reliable) after trialling X API.
- Collection goal: support sentiment and network analysis (May 8 discussion).

## 2. Project Planning Week 2 (May 11 – May 17)

- Data collected and checked for entity references (celebrities/brands).
- Research questions narrowed following tutor consultations on May 12 and 13.
- Preprocessing standardised; entity lists added for matching.
- Notebook structure reworked to separate cleaning, sentiment, and network analysis.

## 3. Project Planning Week 3 (May 18 – May 24)

- **Sentiment Analysis:** Defined requirements, tested VADER/BERT, generated final datasets, and created graphs.
- **Network Analysis:** Developed celebrity co-mention and celebrity-brand networks. As well as community detection, centrality score reporting, and tuned LDA topic modeling.
- **Topic Modelling:** Analysed different topic modelling techniques (LDA vs NMF) and applied the best one to the dataset, analysed each topic further using wordclouds and sentiment analysis.
- Report writing completed, integrating all sections: introduction, data retrieval, problem formation, preprocessing, sentiment analysis, network analysis, topic modelling, conclusion and references.

## 4. Team Contributions

- **Akshita:** Topic shaping, data collection (YouTube comments), applying topic modelling, detailed topic analysis, topic modelling report write-up.
- **Cooper:** Coordination, data collection, entity mapping, preprocessing, sentiment analysis, core report sections.
- **Senglong:** X API exploration, network analysis (graphs, co-mention, centrality, community, LDA), network report write-up.

## 5. Key Meeting Dates

- **May 5:** Topic and direction agreed.
- **May 8:** Data collection process discussed.
- **May 12/13:** Consultations/Feedback on deliverables and topic refinement.
- **May 17:** Group meeting to apply feedback and assign remaining work.
- **May 24:** Final submission checks and document collation.

## 4. Equitable Workload?

- **Akshita:** Yes
- **Senglong:** Yes
- **Cooper:** Yes

# Weekly Timesheets

| Student Name: Akshita Agrawal (s3933878) [Contribution: 33.33%]              |            |            |             |                                                                                                                                                                                            |                      |
|------------------------------------------------------------------------------|------------|------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| WEEK                                                                         | DATE       | TIME START | TIME FINISH | TASK DESCRIPTION                                                                                                                                                                           | TIME TAKEN (in h:mm) |
| 4/05/2026 - 10/05/2026                                                       | 5/5/2026   | 12:10 PM   | 1:10 PM     | Group meeting to discuss topic                                                                                                                                                             | 1:00                 |
|                                                                              | 8/5/2026   | 3:00 PM    | 3:20 PM     | Group meeting to discuss data collection process                                                                                                                                           | 0:20                 |
|                                                                              | 10/5/2026  | 3:00 PM    | 4:00 PM     | Group meeting to discuss data collection                                                                                                                                                   | 1:00                 |
|                                                                              | 10/5/2026  | 4:00 PM    | 6:00 PM     | Fetching youtube comments data for Met Gala 2026                                                                                                                                           | 2:00                 |
| TOTAL TIME SPENT (in h:mm)                                                   |            |            |             |                                                                                                                                                                                            | 4:20                 |
| 11/05/2026 - 17/05/2026                                                      | 11/5/2026  | 3:00 PM    | 3:30 PM     | Group meeting to discuss next steps and rework research questions                                                                                                                          | 0:30                 |
|                                                                              | 12/5/2026  | 10:30 AM   | 11:00 AM    | Attended consultation to get clarification on assignment deliverables                                                                                                                      | 0:30                 |
|                                                                              | 13/5/2026  | 3:00 PM    | 3:30 PM     | Attended consultation to get feedback on chosen topic                                                                                                                                      | 0:30                 |
|                                                                              | 17/5/2026  | 4:15 PM    | 4:45 PM     | Group meeting to discuss given feedback and next steps                                                                                                                                     | 0:30                 |
| TOTAL TIME SPENT (in h:mm)                                                   |            |            |             |                                                                                                                                                                                            | 2:00                 |
| 18/05/2026 - 24/05/2026                                                      | 22/5/2026  | 5:00 PM    | 6:00 PM     | Understanding data and researching topic modelling techniques                                                                                                                              | 1:00                 |
|                                                                              | 23/5/2026  | 8:00 PM    | 10:30 PM    | Testing variants for optimal topic modelling such as bigrams, unigrams, stopwords etc.                                                                                                     | 2:30                 |
|                                                                              | 24/5/2026  | 9:30 AM    | 1:30 PM     | Finalising topic modelling pipeline and running detailed analysis on each topic                                                                                                            | 4:00                 |
|                                                                              | 24/5/2026  | 2:30 PM    | 4:30 PM     | Adding and editing topic modelling approach to report                                                                                                                                      | 2:00                 |
|                                                                              | 24/5/2026  | 5:30 PM    | 6:15 PM     | Group meeting to conduct final submission checks and compile documents                                                                                                                     | 0:45                 |
|                                                                              | 24/5/2026  | 6:30 PM    | 9:30 PM     | Working on report - topic analysis & conclusion & fixing report formatting                                                                                                                 | 3:00                 |
|                                                                              | 24/5/2026  | 9:30 PM    | 10:00 PM    | Fixing project plan and personal statement                                                                                                                                                 | 0:30                 |
|                                                                              | 24/5/2026  | 10:00 PM   | 11:30 PM    | Started presentation slides                                                                                                                                                                | 1:30                 |
| TOTAL TIME SPENT (in h:mm)                                                   |            |            |             |                                                                                                                                                                                            | 15:15                |
| Student Name: Cooper Agustin (s3935261) - Team Leader [Contribution: 33.33%] |            |            |             |                                                                                                                                                                                            |                      |
| WEEK                                                                         | DATE       | TIME START | TIME FINISH | TASK DESCRIPTION                                                                                                                                                                           | TIME TAKEN (in h:mm) |
| 4/05/2026 - 10/05/2026                                                       | 5/5/2026   | 12:10 PM   | 1:10 PM     | Group meeting to discuss topic                                                                                                                                                             | 1:00                 |
|                                                                              | 8/5/2026   | 3:00 PM    | 3:20 PM     | Group meeting to discuss data collection process                                                                                                                                           | 0:20                 |
|                                                                              | 10/5/2026  | 3:00 PM    | 4:00 PM     | Group meeting to discuss data collection                                                                                                                                                   | 1:00                 |
| TOTAL TIME SPENT (in h:mm)                                                   |            |            |             |                                                                                                                                                                                            | 2:20                 |
| 11/05/2026 - 17/05/2026                                                      | 11/5/2026  | 3:00 PM    | 3:30 PM     | Group meeting to discuss next steps and rework research questions                                                                                                                          | 0:30                 |
|                                                                              | 11/5/2026  | 4:00 PM    | 5:30 PM     | Cleaning data collection process + data pre-processing + adding celebrity & brand entities                                                                                                 | 1:30                 |
|                                                                              | 11/5/2026  | 7:00 PM    | 8:00 PM     | Celebrity co-mention and celebrity-brand co-mentions                                                                                                                                       | 1:00                 |
|                                                                              | 12/5/2026  | 10:30 AM   | 11:00 AM    | Attended consultation to get clarification on assignment deliverables                                                                                                                      | 0:30                 |
|                                                                              | 17/5/2026  | 4:15 PM    | 4:45 PM     | Group meeting to discuss given feedback and next steps                                                                                                                                     | 0:30                 |
|                                                                              | 17/5/2026  | 1:00 PM    | 9:00 PM     | Sentiment analysis requirements, preprocessing variants, VADER and BERT sentiment                                                                                                          | 8:00                 |
| TOTAL TIME SPENT (in h:mm)                                                   |            |            |             |                                                                                                                                                                                            | 12:00                |
| 18/05/2026 - 24/05/2026                                                      | 18/5/2026  | 9:00 AM    | 6:00 PM     | Generated updated processed datasets, separated network-analysis work from preprocessing, finalised sentiment analysis                                                                     | 9:00                 |
|                                                                              | 18/5/2026  | 2:00 PM    | 4:00 PM     | Introduction, data retrieval added to report                                                                                                                                               | 2:00                 |
|                                                                              | 21/05/2026 | 9:00 PM    | 9:30 PM     | Reorganised preprocessing, cleaned notebook outputs, sentiment analysis graphs, Problem formation, data collection and limitations, data preprocessing, sentiment analysis added to report | 0:30                 |
|                                                                              | 21/05/2026 |            |             |                                                                                                                                                                                            | 0:00                 |
|                                                                              | 24/5/2026  | 5:30 PM    | 6:15 PM     | Group meeting to conduct final submission checks and compile documents                                                                                                                     | 0:45                 |
| TOTAL TIME SPENT (in h:mm)                                                   |            |            |             |                                                                                                                                                                                            | 12:15                |
| Student Name: Senglong Te (s4054917) [Contribution: 33.33%]                  |            |            |             |                                                                                                                                                                                            |                      |
| WEEK                                                                         | DATE       | TIME START | TIME FINISH | TASK DESCRIPTION                                                                                                                                                                           | TIME TAKEN (in h:mm) |
| 4/05/2026 - 10/05/2026                                                       | 5/5/2026   | 12:10 PM   | 1:10 PM     | Group meeting to discuss topic                                                                                                                                                             | 1:00                 |
|                                                                              | 8/5/2026   | 3:00 PM    | 3:20 PM     | Group meeting to discuss data collection process                                                                                                                                           | 0:20                 |
|                                                                              | 9/5/2026   | 10:00 AM   | 12:00 PM    | Trying to fetch data using X API for Met Gala                                                                                                                                              | 2:00                 |
|                                                                              | 10/5/2026  | 3:00 PM    | 4:00 PM     | Group meeting to discuss data collection                                                                                                                                                   | 1:00                 |
|                                                                              | 10/5/2026  | 7:00 PM    | 8:00 PM     | Built initial graph to visualise data                                                                                                                                                      | 1:00                 |
| TOTAL TIME SPENT (in h:mm)                                                   |            |            |             |                                                                                                                                                                                            | 5:20                 |
| 11/05/2026 - 17/05/2026                                                      | 11/5/2026  | 3:00 PM    | 3:30 PM     | Group meeting to discuss next steps and rework research questions                                                                                                                          | 0:30                 |
|                                                                              | 11/5/2026  | 12:00 PM   | 2:30 PM     | Committed sample graph and built indexes for lookup                                                                                                                                        | 2:30                 |
|                                                                              | 12/5/2026  | 10:30 AM   | 11:00 AM    | Attended consultation to get clarification on assignment deliverables                                                                                                                      | 0:30                 |
|                                                                              | 17/5/2026  | 4:15 PM    | 4:45 PM     | Group meeting to discuss given feedback and next steps                                                                                                                                     | 0:30                 |
| TOTAL TIME SPENT (in h:mm)                                                   |            |            |             |                                                                                                                                                                                            | 4:00                 |
| 18/05/2026 - 24/05/2026                                                      | 19/5/2026  | 12:00 PM   | 4:30 PM     | Building celebrity co-mention and celebrity-brand co-mention network graphs, as well as community detection                                                                                | 4:30                 |
|                                                                              | 20/5/2026  | 11:00 AM   | 12:00 PM    | Fixed centrality score reports + LDA analysis on communities                                                                                                                               | 1:00                 |
|                                                                              | 21/5/2026  | 10:00 AM   | 11:00 AM    | Tuned LDA parameters to get meaningful topics                                                                                                                                              | 1:00                 |
|                                                                              | 22/5/2026  | 3:00 PM    | 6:00 PM     | Adding network analysis section to report                                                                                                                                                  | 3:00                 |
|                                                                              | 24/5/2026  | 5:30 PM    | 6:15 PM     | Group meeting to conduct final submission checks and compile documents                                                                                                                     | 0:45                 |
| TOTAL TIME SPENT (in h:mm)                                                   |            |            |             |                                                                                                                                                                                            | 10:15                |

## **Personal Reflection - Cooper Agustin**

My main contributions to the project were entity mapping, data collection, data exploration, preprocessing, and sentiment analysis using VADER and RoBERTa. A key learning from both SMNA and this project was the importance of having quality data and applying the right preprocessing. After the initial data collection, the sentiment and network analysis did not produce clear or meaningful results. Careful preprocessing was needed to accurately match celebrities and designer brands in our entity set, which made the later sentiment and network analysis much more useful. The project also showed the value of machine learning models such as BERT, especially RoBERTa for Twitter posts. Used alongside VADER, it demonstrated how modern tools can analyse thousands of posts efficiently with only a few hours of work. The project highlighted how simple but deliberate steps in data collection and preprocessing can make a large dataset far more useful for analysis.

## **Personal Reflection - Akshita Agrawal**

I contributed primarily to the topic modelling and topic analysis components of the project. I initially proposed using the Met Gala as the project topic and carried out the early stages of YouTube data collection. My main responsibility was identifying the most suitable topic modelling approach for our data (LDA or NMF), evaluating models based on coherence scores, analysing topic quality through example comments and selecting the final model used for subsequent analysis. I also implemented the detailed topic analysis, including topic distributions, word clouds and sentiment integration. One of the main challenges I found was working with short and noisy social media comments, which often made topic separation difficult. To improve topic quality, I experimented with preprocessing techniques such as stopword selection, bigrams, TF-IDF vectorisation, and different topic counts. This process helped me develop a stronger understanding of practical NLP workflows and topic modelling techniques. In terms of team contribution, I believe our team worked very well overall in dividing responsibilities while still regularly discussing results and making decisions together.

## **Personal Reflection - Senglong Te**

My main contributions to the project included initial data collection research i.e. researching different API(s) to use for data collection, and network analysis measures. In terms of API, I researched X api, Reddit, Threads, Meta's graphAPI, lastly Youtube API. X api was found to be far too expensive: basic tweet extraction + author information would have cost nearly 100USD for 10000 comments beyond the scope of the assignment. Reddit/Meta required us to request permission. We ended up settling on Youtube's API. After discussion of possibly using celebrity as nodes and co-mentions as node edges. I set out to determine if this structure would return interpretable and interesting results. The graph turned out to be very rich in information. From running basic centrality analysis revealed some very interesting results; Who are central figures to the met-gala? Which celebrity nodes helped bridge different communities. I then went further trying to understand the nature of the communities in the met-gala. This was done by running community analysis on the graph. From this I discovered communities clustered around different pop cultures from different countries. I was then interested in seeing what the conversation in these communities were like. So I ran topic analysis on each community. Initially the topics weren't coherent but after tuning LDA's parameters eventually I did arrive at interesting coherent topics.